INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY
HYDERABAD

5-Bit CLA Adder

VLSI-Design Course Project

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Abstract—This report presents the design and implementation of a 5-bit synchronous Carry Look -Ahead (CLA) adder using 180 nm CMOS technology. The CLA architecture minimizes carry propagation delay by computing carry signals in parallel through propagate and generate logic.

I) Introduction

A 5-bit Carry Look-Ahead (CLA) Adder is a high-performance digital arithmetic circuit designed to add two 5-bit binary numbers with significantly reduced propagation delay compared to a conventional Ripple Carry Adder (RCA). Instead of waiting for each carry to ripple through successive full adders, the CLA architecture computes the carries in parallel using propagate (P) and generate (G) signals, enabling much faster addition.

In this project, the CLA adder is implemented using 180 nm CMOS technology, with all blocks designed at the transistor level. The architecture consists of a Propagate/Generate block, a Carry Look-Ahead network, and a Sum computation block. The design is made synchronous using D Flip-Flops (DFFs). All modules were verified using NGSPICE, followed by layout and post-layout validation in MAGIC.

II) Proposed Design

The proposed design uses a static CMOS implementation of a 5-bit carry lookahead adder. The overall structure is divided into three main functional blocks, along with TSPC D flip-flop registers that provide clocked synchronization between stages. The first block is the PG block, which generates the propagate and generate signals for each bit. The second block is the CLA block, which computes the carry values using the propagate and generate information in a fast parallel manner. The third block is the Sum block, which produces the final sum outputs based on the calculated carries. This modular organization improves speed, supports clean pipelining through the TSPC registers, and ensures reliable operation under synchronous control.

We are to add two 5-bit numbers, namely A4 A3 A2 A1 A0 and B4 B3 B2 B1 B0. The initial carry input C0 is set to 0. The design first computes the propagate signals Pi and the generate signals Gi for each bit position. These signals indicate whether a bit pair will pass an incoming carry or generate a new carry. The CLA block then uses these Pi and Gi values to determine all carry outputs C1 to C5 in parallel, which significantly reduces delay compared to a ripple carry structure. After the carry values are available, the Sum block produces the final sum outputs S4 to S0 by combining the propagate signals with the corresponding carry inputs. The use of TSPC D flip-flop registers at key boundaries ensures proper synchronization, allowing each stage to operate reliably under clocked control. This approach achieves high speed while maintaining a simple, static CMOS logic style.

1) Propagate and Generate Block:-

For each bit position i, the inputs are Ai and Bi. Using these inputs, the propagate signal Pi and the generate signal Gi are defined as:
Pi = Ai XOR Bi
Gi = Ai AND Bi
These are used to compute carry out in the CLA block, when Gi = 1, a carry is generated, irrespective of the carry in. Whereas, is Pi = 1, and Gi=0, then the carry in is propagated to carry out, i.e., Ci+1=Ci.

2) Carry-Lookahead Block:-

The carry relationship between consecutive stages is expressed as:
Ci+1 = Gi + (Pi . Ci)
This means that a carry is generated either directly by Gi or passed forward through Pi from the previous carry Ci.

3) Sum Block:-

The sum output for each bit is calculated using the corresponding inputs and carry value. The standard sum equation is:

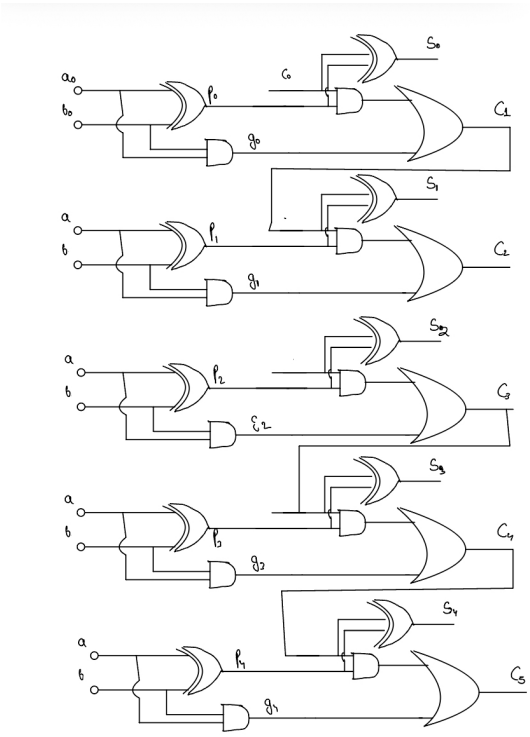
Si = Ai XOR Bi XOR Ci

Since Ai XOR Bi is equal to Pi, an equivalent expression for the sum is:

Si = Pi XOR Ci

This shows that once the propagate signals and carry values are known, the sum bits can be produced with minimal delay, this reduces delay, as the carry out is quickly rippled through, as soon as carry is inputted. This applied on two 5-bit numbers is a 5-bit CLA Adder.

Diagram of Static Implementation of 5-bit CLA



III) Design of Adder Modules and D Flip Flop

D-Flip Flop:-

The D flip-flop used in the design is based on the True Single Phase Clock (TSPC) topology. The TSPC D flip-flop operates through three sequential stages controlled by a single clock signal. In the first stage, the input D is sampled while the clock is low, and this value is isolated once the clock transitions high. The second stage processes and stores the sampled value during the high phase of the clock, maintaining stability when the clock returns low. The final stage drives the output Q, where the node is pre-charged in the low phase and evaluated during the high phase. This arrangement results in an output transition that effectively occurs on the rising edge of the clock, enabling edge-triggered behavior without the need for complementary clock phases.
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PG Block:-

The propagate and generate signals are produced using static logic. The propagate signal Pi is generated using a transmission-gate XOR gate, while the generate signal Gi is obtained using a static AND function built from a NAND gate followed by an inverter. All gates use CMOS static logic other than XOR, and the sizing follows the uniform ratio of PMOS width equal to twice the NMOS width. This keeps the rise and fall times balanced across all operation since the resistance is equalized to an idealized inverter.

Carry Lookahead Block:-

The carry lookahead block is implemented using static CMOS gates such as NAND, NOR, AND, and OR. The carry equations are mapped into two-input gates to maintain simplicity and reduce delay. All devices follow the same sizing rule of PMOS being twice the width of NMOS, ensuring consistent electrical behavior. Minor upsizing may be applied to the final carry gates to improve the speed of the critical path without changing the overall ratio.

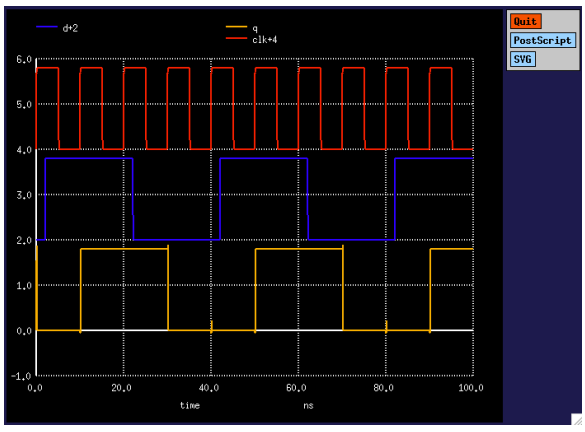
Sum Logic Block:-

The sum bits are produced using a second XOR operation between the propagate signal and the corresponding carry value. This XOR is again implemented using transmission-gate logic followed by an inverter to restore signal levels. The transistors in this block also follow the uniform two-to-one PMOS-to-NMOS sizing rule, which simplifies design and maintains stable switching characteristics.

IV) Simulation and Layout

All submodules were simulated in NGSPICE to verify their functional behavior. The layouts for every block were designed in MAGIC using the SCN6M_DEEP.09.tech27 technology file. Post-layout simulations were performed to confirm that the circuits met the required timing and that their functionality remained correct after parasitic extraction.

NGSpice simulation of D-Flip Flop:



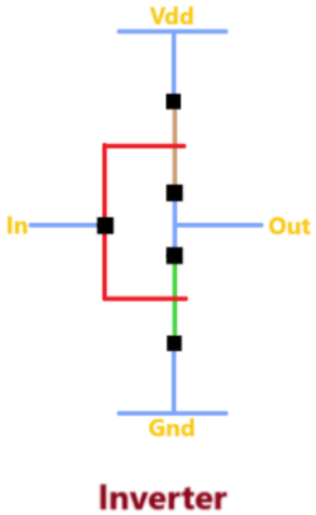
We also find the timing constraints of the D-Flip Flop through NGSpice by measuring the delay between certain nodes in Flip - Flop.

here is the setup time, hold time and tcqfor the important timing constraints of the D - Flip Flop Simulation, where tsu= setup time; th=hold time and tcq_rise =T clk to q.

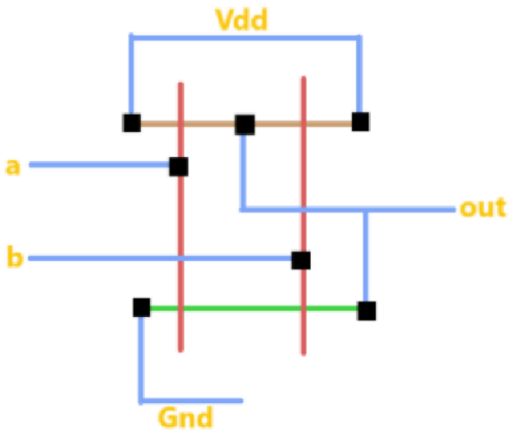
```
thfall          =  5.565845e-11 targ=  1.056585e-10 trig=  5.000000e-11
thrise          =  5.011825e-11 targ=  5.200118e-09 trig=  5.150000e-09
th              =  5.28884e-11
tsufall         =  2.495720e-11 targ=  2.074957e-09 trig=  2.050000e-09
tsurise         =  3.053864e-09 targ=  2.520386e-08 trig=  2.215000e-08
tsu             =  1.53941e-09

t_rise          =  9.164557e-11 targ=  9.842237e-11 trig=  6.776795e-12
t_fall          =  2.613691e-11 targ=  1.884370e-10 trig=  1.623000e-10
tpd_ave         =  5.88912e-11
tcq_rise        =  2.564727e-11 targ=  7.564727e-11 trig=  5.000000e-11
```

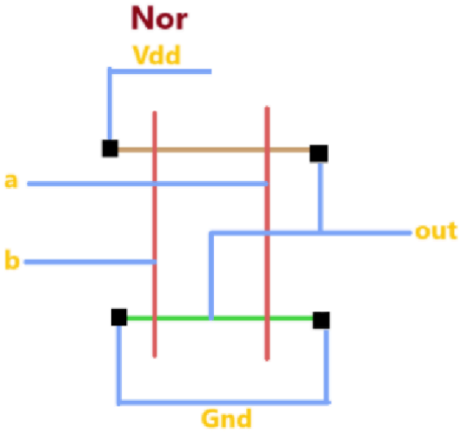
Stick Diagrams:-

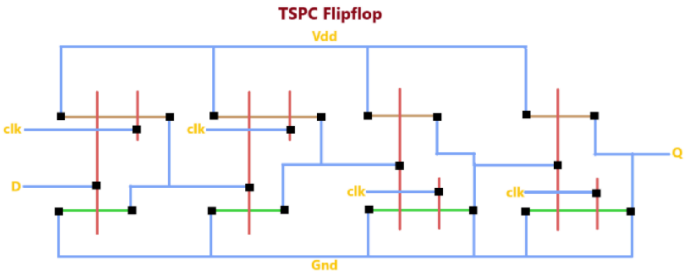


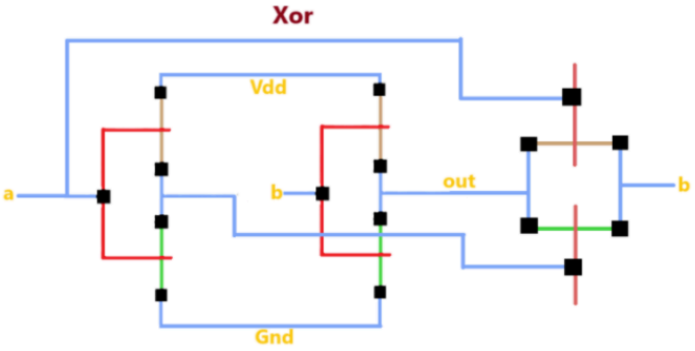
Inverter



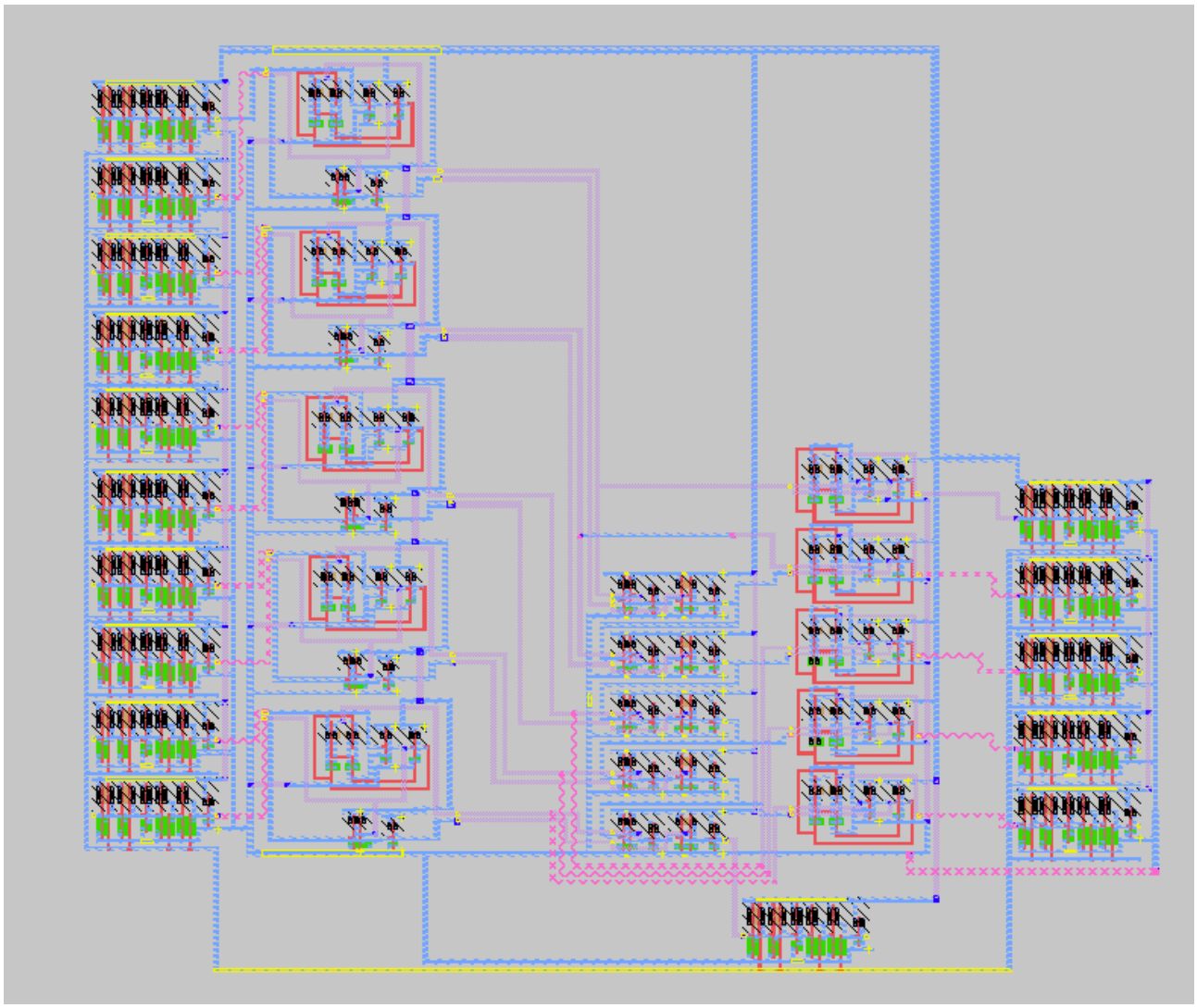
Nand



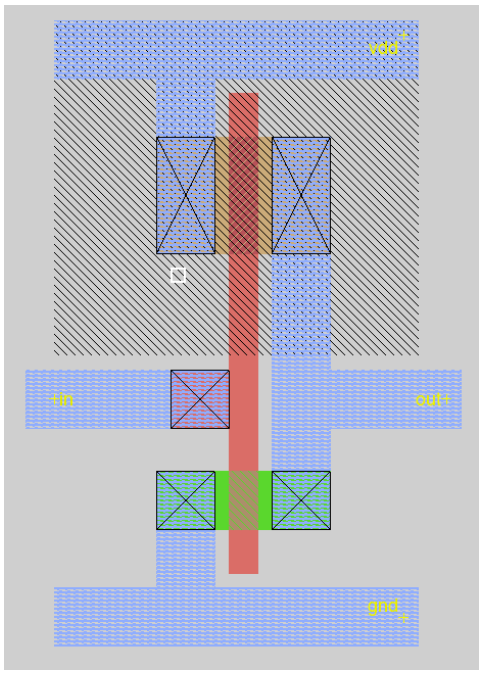




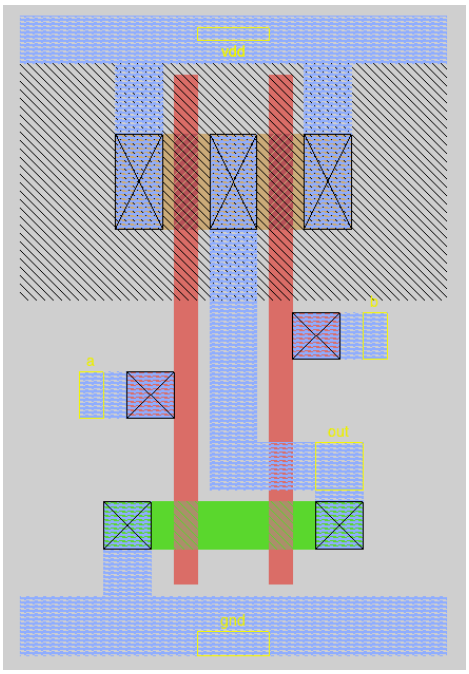
MAGIC Implementation and Post Layout Testing:-



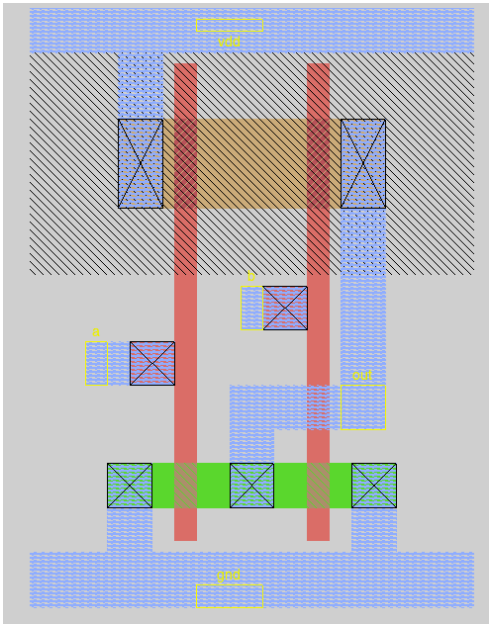
Final Layout of the Complete 5-bit CLA adder.



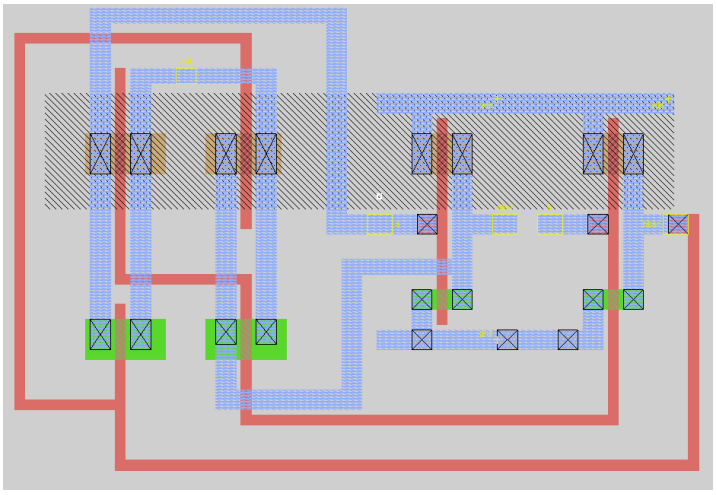
Inverter Implementation



NAND Gate

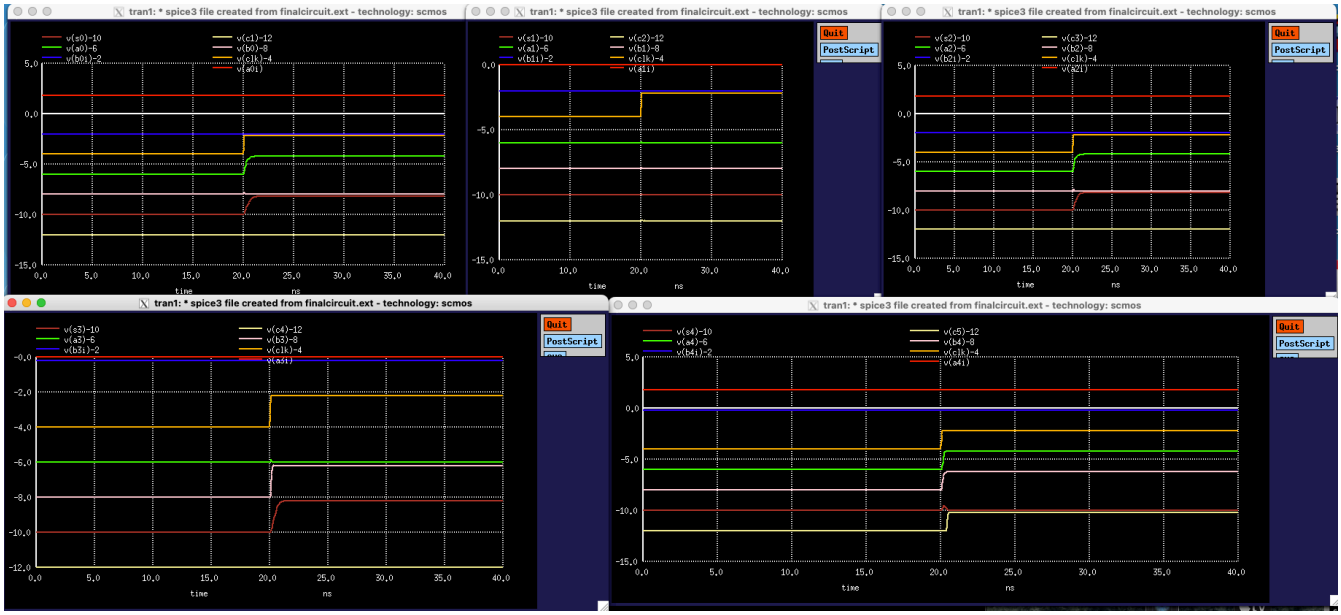


NOR gate

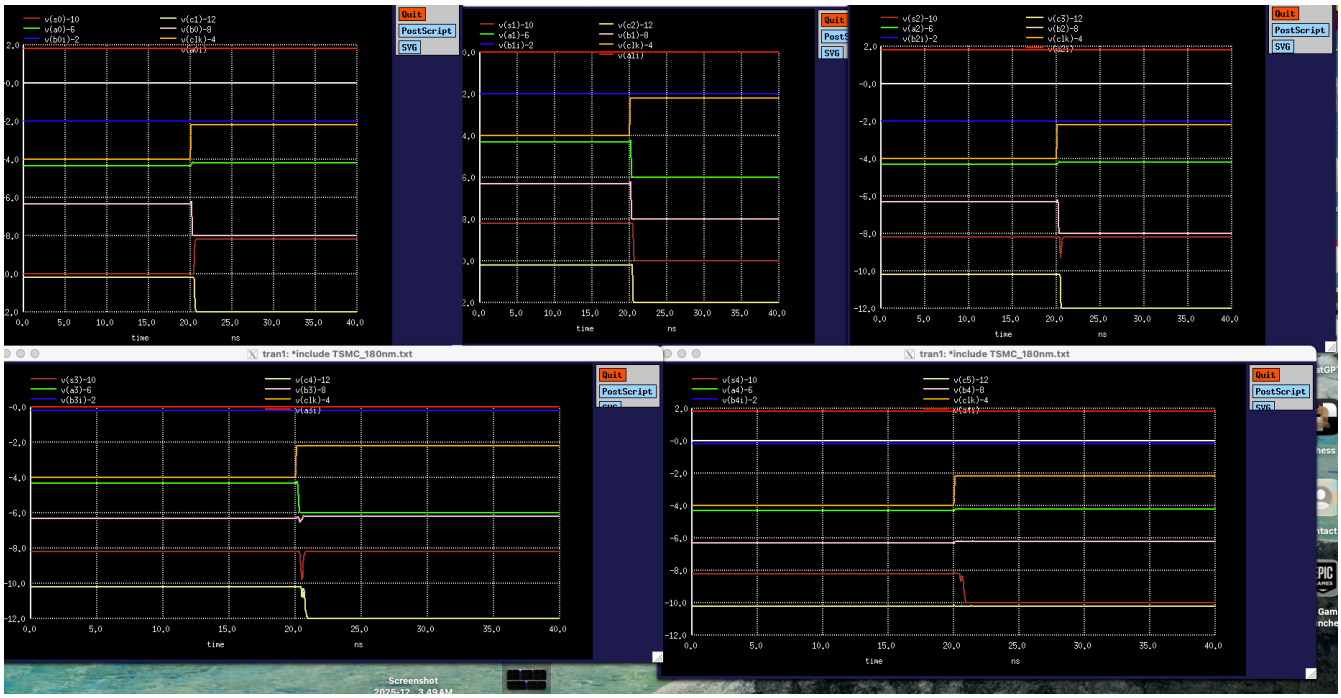


XOR gate

Simulating the addition of A=10101 and B=11000;



Post Layout simulation



Pre Layout simulation

Here, we present the sum bits S3 and S4 for both pre-layout and post-layout simulations. The sum outputs match in both cases, confirming that the functional behavior of the adder remains consistent before and after layout extraction.

```
tpd_rise            =  3.255581e-10 targ=  2.037556e-08 trig=  2.005000e-08
```

```
tpd_rise            =  4.148924e-10
```

The delay between the clock and S0 is measured as 325.558 ps in the pre-layout simulation and 414.892 ps in the post-layout simulation. Although post-layout delays are typically larger, this difference is explained by changes in loading and actual routing after layout. In the extracted design, some nets experience reduced effective capacitance or shorter interconnect paths compared to the idealized pre-layout model. As a result, the post-layout combinational path from the flip-flop output through the adder logic to S0 becomes slightly faster. Small variations in the clock arrival time at the flip-flops and more accurate modeling of parasitics can also shift the measured edge, giving a lower delay in the post-layout case.

Pre-Layout Simulation	Post-Layout Simulation
We can see from the plots and from the measurements that there is less delay in the pre layout simulation because of lack of parasitic capacitances and skews. The S0 plot has less curvature compared to the one the post layout simulation has.	In the post-layout simulation, the extracted parasitic resistances and capacitances from the actual layout are included, resulting in more realistic timing behavior. These parasitics introduce additional loading and influence signal propagation, causing the S0 waveform to show more curvature and a smoother transition compared to the pre-layout case.

Minimum Clock Period Calculation

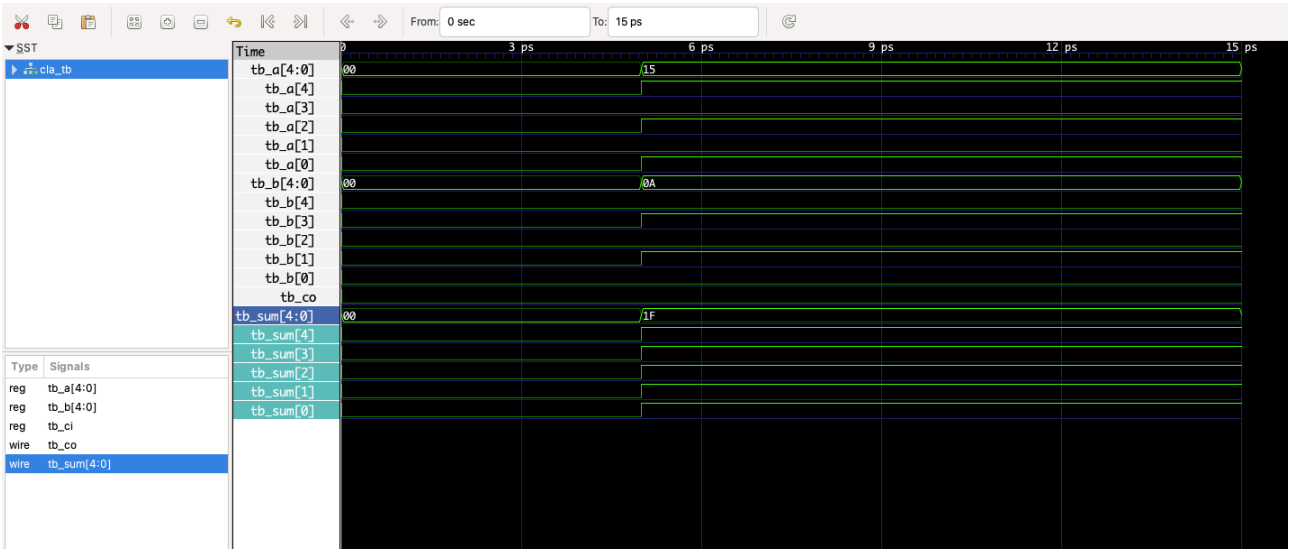
For the flip-flop to operate correctly, the clock period must satisfy the following condition:
TCLK_min = tCQ + tlogic_max + tsetup
where tlogic_max represents the maximum combinational logic delay between two sequential elements.
For the 5-bit CLA adder pipeline, this becomes:
TCLK_min = tCQ_DFF + tCLA_max + tsetup_DFF
This ensures that data launched by the first flip-flop reaches the input of the second flip-flop and meets its setup time requirement before the next rising clock edge.

Maximum Operating Frequency

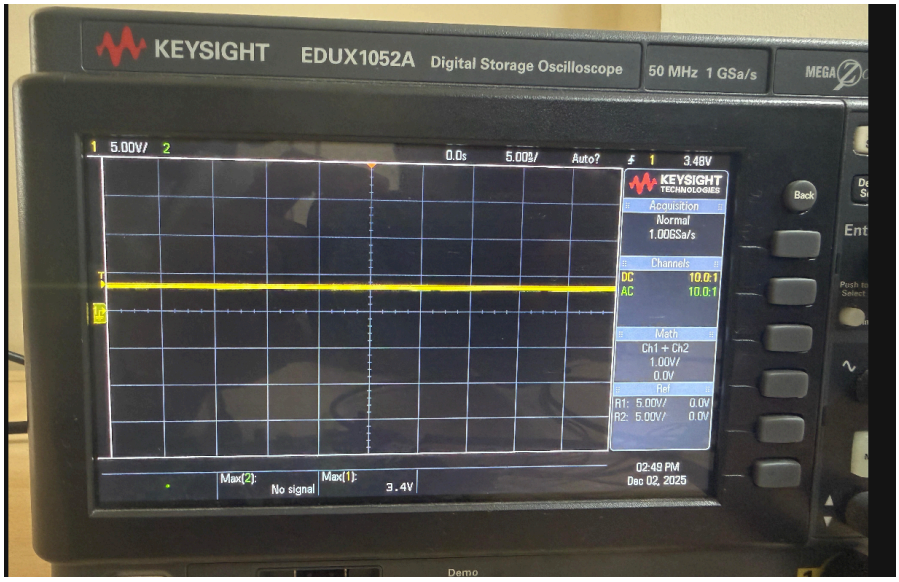
The highest allowable clock frequency is obtained by taking the reciprocal of the minimum clock period:
fmax = 1 / TCLK_min
= 1 / (tCQ + tCLA_max + tsetup)
This expression sets the upper bound on the working frequency of the pipelined CLA adder, determined by the combined delay of the launching flip-flop, the worst-case delay of the CLA combinational block, and the setup time of the receiving flip-flop.

Verilog Simulations:-

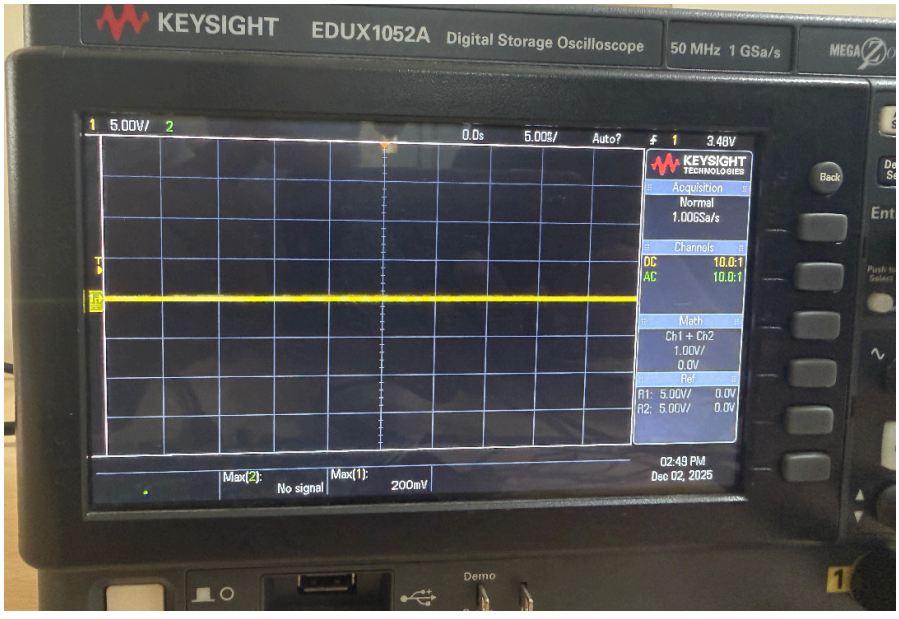
GTKWave Simulation of addition of A = 10101 and B=01010
C0=0 always
Sum=11111 and C5=0;



FPGA Oscilloscope Simulations



For S4=1



For C5=0

Acknowledgement:-

I would like to express my sincere gratitude to Prof. Abhishek Srivastava for his guidance, encouragement, and the opportunity to work on this VLSI design project. The course provided a strong foundation in CMOS VLSI concepts, and I truly enjoyed applying the theoretical knowledge to practical circuit implementation.

I would also like to thank the Teaching Assistants, for their continuous support, timely feedback, and help during the labs and project development. Their explanations and insights greatly enhanced my understanding of the tools and design methodologies.

References:-

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- Jiren Yuan & C. Svensson, “New single-clock CMOS latches and flip-flops with improved speed and power savings,” IEEE Journal of Solid-State Circuits, 1997.
- Yu-Ting Pai & Yu-Kumg Chen, "The fastest carry lookahead adder"